Anomaly Detection Results

Baseline Results

Metric	Value
Total data points	1050
Anomalies injected	50
Anomalies detected by the model	53
True positives (correctly flagged)	39
False positives (normal data flagged)	14
False negatives (anomalies missed)	11
Precision	0.736
Recall	0.780
F1 Score	0.757

Scatter Plot Observations

Based off the 2 Scatter Plot charts, the left chart shows pin points of real anomalies and a cluster of normal data, the right chart shows where the model is predicting the anomalies to be.

Parameter Experiment

changed `contamination` from `0.05` to `0.1`.

Metric	Default (0.05)	Changed (0.1)

Anomalies detected	53	105
True positives	39	40
False positives	14	65
Precision	0.736	0.381
Recall	0.780	0.800
F1 Score	0.757	0.516

explaining what changed when increasing contamination:

Increasing the contamination caused our model to display more anomalies giving us lower accuracy.

Hyperparameter Experiments

Contamination Sweep

Contamination	**Anomalies Detected**	**Precision**	**Recall**	**F1 Score**
0.01	11	1.000	0.640	0.780
0.03	32	1.000	0.640	0.780
0.05	53	0.736	0.780	0.757
0.1	105	0.736	0.780	0.757
0.2	210	0.190	0.800	0.308

Which contamination value gives the best F1 score?

Best results given were Contamination 0.03, This gives us a F1 Score of 0.780 (78%)

Tree Count Experiment

n_estimators	Precision	Recall	F1 Score	Training Time (sec)
50	0.698	0.740	0.718	0.632s
100	0.736	0.780	0.757	0.894s
200	0.717	0.760	0.738	1.208s
500	0.717	0.760	0.738	3.271s

At what point do more trees stop improving the results?

A count of 200 stops showing improvements and only adds training time.

Does training time increase linearly with the number of trees?

Your answer here

Sample Size Experiment

max_samples	Precision	Recall	F1 Score
auto	0.736	0.780	0.757
64	0.585	0.620	0.602
128	0.698	0.740	0.718
256	0.736	0.780	0.757

Which sample size gives the best F1 score?

Auto or 256 gives the best F1 scores as they both are 75%

Best Configuration

Configuration	Precision	Recall	F1 Score
Default model	1.000	0.220	0.361
Your best model	0.736	0.780	0.757
Improvement	-0.264	0.560	0.397

analyzing the difference:

Looking at the metrics from our default results going to our best results there is a huge difference, losing 26% on precision is fine because of the tradeoff with gaining 56% on recalls and 39% on f1 scores which will give us more stable results for our anomaly detection.

Model Comparison

LOF Default Results

Metric	LOF Default
Anomalies detected	40
Precision	0.755
Recall	0.800
F1 Score	0.777

LOF Tuning (n_neighbors)

n_neighbors	Precision	Recall	F1 Score
10	0.755	0.800	0.777

20	0.755	0.800	0.777
50	0.755	0.800	0.777

Which n_neighbors value gives the best F1 score?

10 n_neighbors is the best when it comes to F1 scoring, precision, recall, and F1 are all the same but it comes down to the time for each n_neighbors, 10 had the fastest time with 0.017s

Head-to-Head Comparison

Metric	**Isolation Forest (Best)**	**LOF (Best)**	**Winner**
Precision	0.736	0.755	LOF
Recall	0.780	0.800	LOF
F1 Score	0.757	0.777	LOF
Training Time	0.359s	0.017s	LOF
Can score new data one at a time?	Yes	No	IF
Needs full dataset present?	No	Yes	IF

Recommendation

Which model should the Grand Marina deploy for real-time anomaly detection on their water system, and why? Write 3-5 sentences.

For the Grand Marina the best model to train would be IF, the main reason is for real-time anomaly detection. With IF you can allow new messages to be scored without needing all of the dataset present, LOF might have faster and more accurate detections but it needs *the whole dataset so it's not valid in this case. IF is also a lot easier to integrate into an existing subscriber, it's only downside is it might give out slightly more false alarms compared to a LOF, but that is manageable with a good security team.*

Optional: One-Class SVM Results

Metric	**One-Class SVM**
Anomalies detected	33
Precision	0.611
Recall	0.660
F1 Score	0.635
Training Time	0.009s

Importing AI Model and Running Attacks Results:

Attack Type	***Rules Catch It?***	***AI Catches It?***	***Dashboard Color***
Fake HMAC (injected message)	*Yes — blocked*	*N/A (never reaches AI)*	*Red*
Replayed message	*Yes — blocked*	*N/A (never reaches AI)*	*Red*
High or low pressure	*No — valid HMAC/timestamp/sequence*	*Yes — anomalous pattern*	*Orange*
Flow surge or drop	*No — values look structurally valid*	*Yes — outside learned range*	*Orange*
Normal sensor data	*Passes*	*Normal*	*Green*